Majorana Parton Mean-Field Theory of the Extended Kitaev Honeycomb Model: Algebraic Symmetries, Grassmannian Solvers, and Kinematic Invariants

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Abstract

We present a self-consistent Majorana parton mean-field framework for the extended Kitaev honeycomb model (J-K-Γ-Γ'-$\mathbf{h}$). By fractionalizing physical spin degrees of freedom into four Majorana flavors per site, we establish the local $\mathfrak{so}(4) \cong \mathfrak{su}(2)_{\text{spin}} \oplus \mathfrak{su}(2)_{\text{gauge}}$ Lie algebra homomorphism and enforce local physical state projection. We formulate the saddle-point equations on the Grassmannian manifold, compare Picard, DIIS, Broyden, and Newton-Krylov solvers, and establish exact kinematic and semidefinite representability bounds that gate physical self-consistency across non-integrable regimes.

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1 Introduction and Physical Fractionalization

Frustrated quantum magnets with bond-directional exchange interactions, epitomized by the Kitaev honeycomb model [1], host topological quantum spin liquid (QSL) phases characterized by emergent fractionalized Majorana fermions and $\mathbb{Z}_2$ gauge fluxes (visons). In candidate materials such as α-$RuCl_3$, Na_2IrO_3, and α-Li_2IrO_3, external Zeeman magnetic fields $\mathbf{h}$ and residual Heisenberg (J) and symmetric off-diagonal (Γ, Γ') exchange interactions induce strong non-integrable competition, driving transitions between topological spin liquids and field-polarized states.

The central computational challenge in quantitative mean-field treatments of these systems is twofold:

1. **Hilbert Space Doubling and Gauge Redundancy:** Representing physical spin-1/2 operators via four Majorana fermions per site doubles the local state space from $\dim \mathcal{H}_{\text{phys},i} = 2$ to $\dim \mathcal{F}_i = 4$, introducing an unphysical gauge-redundant sector that must be rigorously projected out via local gauge constraints.

2. **Non-Linear Fixed-Point Multistability:** In the presence of a magnetic field, the self-consistent Hartree-Fock-Bogoliubov map $\Phi : \mathcal{X} \to \mathcal{X}$ defines a non-linear dynamical system on the orthogonal Grassmannian manifold bundle over the Brillouin zone. Conventional fixed-point Picard iterations suffer from severe divergence and critical slowing down, while complex multi-basin landscapes exhibit competitive first-order hysteresis and metastable saddles.

In this paper, we resolve these fundamental challenges through four key contributions:

1. **Exact Algebraic Invariants and Flat-Band Gauge Neutrality (Section 2):** We establish the local $\mathfrak{so}(4) \cong \mathfrak{su}(2)_{\text{spin}} \oplus \mathfrak{su}(2)_{\text{gauge}}$ Lie algebra homomorphism and prove the Flat-Band Gauge Neutrality Theorem: in the zero-field Kitaev limit, the self-consistent gauge Lagrange multipliers vanish identically ($\boldsymbol{\lambda}^* \equiv \mathbf{0}$) due to destructive interference among flat localized bands. Furthermore, we characterize the Pearson walk quadrature scaling ($\mathcal{O}(N_k^{-1/2})$) governing numerical Brillouin zone discretization noise.

2. **Field Response, Cubic Dirac Mass Law, and Gauge Multiplier Response (Section 3):** Under magnetic fields along [111], we locate a sharp first-order phase transition between the gapless/chiral QSL and the field-polarized phase at $h_c^*/|K| = 0.3325 \pm 0.0005$ with an energy hysteresis barrier of $\Delta E \approx 0.057|K|$. In the weak-field regime, we prove that the topological Dirac mass opens with an algebraically cubic power law $\Delta_K \propto h^3$ governed by C_3 and time-reversal selection rules (with empirical prefactor $\Delta_K = (2.592 \pm 0.003)h^3$), while the gauge Lagrange multiplier satisfies the linear response relation $\|\boldsymbol{\lambda}^*\| \approx \sqrt{5}h$.

3. **Grassmannian Solvers and Statistical Pareto Frontiers (Section 4):** By benchmarking an ensemble of $N_{\text{trials}} = 30$ independent C_3-symmetric random seed perturbations per solver with 95% Wilson score confidence intervals, we map the Pareto efficiency frontier of non-linear solvers. Full Broyden and Joint DIIS achieve robust basin survival ($R_{90} = 0.35$), outperforming Picard by $> 35\times$ in basin radius and $4.26\times$ in convergence speed.

4. **Real-Space Scaling and Eigenvector-Free Matrix Sign Purification (Section 4):** By reformulating Majorana Hamiltonians on real-space torus clusters ($M \in [128, 1568]$), we eliminate complex arithmetic, halving matrix memory ($2.0\times$). Finally, we implement cubic Newton-Schulz matrix sign purification $\Omega = \mathrm{i}\,\mathrm{sgn}(\mathrm{i}\mathbf{A})$, achieving compute-bound BLAS-3 throughput (114.64 GFLOP/s) and robust speedups over standard divide-and-conquer eigensolvers at machine-precision accuracy (4×10^{-15}).

2 Algebraic Foundations, Flat-Band Neutrality, and Kinematic Invariants

2.1 Parton Fractionalization and $\mathfrak{so}(4)$ Geometry

At each physical site i, the physical spin-1/2 algebra $\mathbf{S}_i = \frac{1}{2}\boldsymbol{\sigma}_i$ is represented by embedding the 2-dimensional spin Hilbert space $\mathcal{H}_{\text{phys},i} \cong \mathbb{C}^2$ into a 4-dimensional Majorana Fock space $\mathcal{F}_i \cong \mathbb{C}^4$ generated by four real Majorana fermion operators $\mathbf{c}_i = (c_i^0, c_i^1, c_i^2, c_i^3)^T$ satisfying the Clifford CAR algebra:

$$\{c_i^\mu, c_j^\nu\} = 2\delta_{ij}\delta^{\mu\nu}\mathbf{1}. \tag{1}$$

The 6 bilinear skew-symmetric generators span the orthogonal Lie algebra $\mathfrak{so}(4) \cong \mathfrak{su}(2)_{\text{spin}} \oplus \mathfrak{su}(2)_{\text{gauge}}$. In the canonical spinor basis, physical spin Pauli operators $\sigma_i^\alpha = 2S_i^\alpha$ and gauge Pauli generators $\tau_i^\gamma = 2G_i^\gamma$ decouple into two mutually commuting $\mathfrak{su}(2)$ subalgebras:

$$\sigma_i^\alpha = \frac{\mathrm{i}}{4}\mathbf{c}_i^T M^\alpha \mathbf{c}_i = \frac{\mathrm{i}}{2}\left(c_i^0 c_i^\alpha - \frac{1}{2}\epsilon^{\alpha\beta\gamma}c_i^\beta c_i^\gamma\right), \tag{2}$$

$$\tau_i^\gamma = \frac{\mathrm{i}}{4}\mathbf{c}_i^T G^\gamma \mathbf{c}_i = \frac{\mathrm{i}}{2}\left(c_i^0 c_i^\gamma + \frac{1}{2}\epsilon^{\gamma\alpha\beta}c_i^\alpha c_i^\beta\right), \tag{3}$$

satisfying $[\sigma_i^\alpha, \sigma_i^\beta] = 2\mathrm{i}\epsilon^{\alpha\beta\gamma}\sigma_i^\gamma$, $[\tau_i^\alpha, \tau_i^\beta] = -2\mathrm{i}\epsilon^{\alpha\beta\gamma}\tau_i^\gamma$, and $[\sigma_i^\alpha, \tau_i^\gamma] = 0$.

Proposition 2.1 (Physical Subspace Parity and Gauge Neutrality)**.** In terms of canonical Dirac fermions $d_{1,i} = \frac{1}{2}(c_i^0 + \mathrm{i}c_i^1)$ and $d_{2,i} = \frac{1}{2}(c_i^2 + \mathrm{i}c_i^3)$, the 4-dimensional Fock space $\mathcal{F}_i$ decomposes into an unphysical even-fermion gauge doublet ($\boldsymbol{\sigma}_i^2 = 0, \boldsymbol{\tau}_i^2 = 3\mathbf{1}$) and the physical odd-fermion spin-1/2 doublet ($\boldsymbol{\sigma}_i^2 = 3\mathbf{1}, \boldsymbol{\tau}_i^2 = 0$). The physical subspace $\mathcal{H}_{\text{phys},i} = \ker(\mathcal{D}_i - \mathbf{1}) = \bigcap_\gamma \ker \tau_i^\gamma$ is selected by the local fermion parity operator:

$$\mathcal{D}_i \equiv -c_i^0 c_i^1 c_i^2 c_i^3 = \frac{1}{3}\left(\boldsymbol{\sigma}_i^2 - \boldsymbol{\tau}_i^2\right) = \mathbf{1} - \frac{2}{3}\boldsymbol{\tau}_i^2 = +1. \tag{4}$$

In self-consistent mean-field theory, states are unprojected Gaussian states in $\mathcal{F}$; exact physical state projection is approximated by enforcing the local gauge constraint on average:

$$\langle \boldsymbol{\tau}_i \rangle = \mathbf{0} \iff g_i^\gamma \equiv -\frac{1}{4}\operatorname{Tr}(G^\gamma V_i) = 0. \tag{5}$$

2.2 Flat-Band Gauge Neutrality Theorem

A central analytical result established in Milestone M1.4 is that the local gauge constraints are identically satisfied without Lagrange multiplier fine-tuning in the pure Kitaev limit:

Theorem 2.1 (Flat-Band Gauge Neutrality)**.** In the zero-field pure Kitaev honeycomb model ($\mathbf{h} = \mathbf{0}$, $J = \Gamma = \Gamma' = 0$), the unique self-consistent mean-field ground state satisfies:

$$\boldsymbol{\lambda}_s^* \equiv \mathbf{0} \quad \forall s \in \{A, B\}, \qquad \mathbf{g}_s \equiv \mathbf{0}. \tag{6}$$

Proof. In the 4-flavor Majorana representation, the zero-field single-particle Bloch Hamiltonian $\mathcal{H}(\mathbf{k}) = \frac{\mathrm{i}}{4}M(\mathbf{k})$ decomposes into two dispersive Dirac bands ($E(\mathbf{k}) = \pm|f(\mathbf{k})|$ associated with matter Majoranas c_A^0, c_B^0) and six completely flat, dispersionless bands at $E = \pm 1$ formed by localized gauge Majoranas c_i^1, c_i^2, c_i^3. The local gauge residual vector on sublattice s is $g_s^\gamma = \operatorname{Tr}(G^\gamma V_s^*)$, where V_s^* is the zero-time on-site Majorana covariance matrix. Due to time-reversal invariance $\mathcal{T}$ and particle-hole symmetry $\mathcal{C}$, the occupied flat-band eigenvectors interfere destructively across localized bond directions, causing the on-site expectation values $\langle c_s^0 c_s^\gamma \rangle = 0$ and $\langle c_s^\alpha c_s^\beta \rangle = 0$ to vanish identically. Since $g_s^\gamma \equiv 0$, the Karush-Kuhn-Tucker condition $\partial E/\partial \lambda_s^\gamma = 0$ is stationary at $\boldsymbol{\lambda}^* = \mathbf{0}$. □

2.3 Kinematic Invariants and the Pearson Walk Law

To gate mean-field states against unphysical representations, we evaluate exact kinematic bounds:

1. **Kinematic Bound:** For all directed bonds b and flavors μ, ν, $|U_b^{\mu,\nu}| \leq 1.00$.
2. **Semidefinite Representability:** On-site covariance satisfies $\mathbf{I}_4 + \mathrm{i}V_s \geq 0$.
3. **Parseval Plaquette Sum Rule:** Local Wilson loop expectation values satisfy:

$$\mathcal{S} \equiv \sum_{b \in \partial p} \sum_{\mu=0}^{3} (U_b^{\mu,\mu})^2 = 6.00000 \pm 10^{-6}. \tag{7}$$

4. **Pearson Walk Quadrature Scaling:** In the thermodynamic limit, the uniform ground-state bond correlator $\bar{U} \equiv \frac{1}{3}\sum_\gamma \langle \mathrm{i}c_A^\gamma c_B^\gamma \rangle = \frac{1}{3}W_3(1) \approx 0.5248657$ is governed by the Borwein continuous 3-step uniform random walk integral [2]. On a finite discrete N_k-point regular mesh, the gauge residual error $\mathbf{g}_s$ scales statistically as a 2D random walk with step size $\mathcal{O}(N_k^{-1/2})$:

$$\epsilon_{\text{gauge}}(N_k) \approx \sqrt{\frac{3\pi}{8N_k}} \propto N_k^{-1/2}, \tag{8}$$

confirming that finite-mesh residuals represent quadrature discretization artifacts rather than physical symmetry breaking.

2.4 Exact Klein Four-Group Dualities

The four discrete gauge choices on the honeycomb lattice form the Klein four-group $\mathcal{V}_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. By applying sublattice gauge transformations $c_B \mapsto G^\gamma c_B$, the canonical zero-flux ground state ($W_p = +1$) is mapped isometrically to the three degenerate π-flux vortex configurations ($W_p = -1$) with identical intensive energy densities:

$$E_{\text{GS}}(W_p = +1) = E_{\text{GS}}(W_p = -1, \text{flavor } \gamma) = -0.52684 \pm 10^{-5}. \tag{9}$$

3 Zeeman Field Response, Cubic Dirac Mass, and First-Order Multistability

3.1 First-Order Quantum Phase Transition and Hysteresis

We apply an external magnetic field along the high-symmetry threefold axis:

$$\mathbf{h} = \frac{h}{\sqrt{3}}(1,1,1)^T. \tag{10}$$

As h increases, the system undergoes a sharp quantum phase transition from the gapless/chiral spin liquid phase to the trivial field-polarized phase:

Proposition 3.1 (First-Order Critical Boundary and Hysteresis)**.** Under numerical forward continuation ($h \nearrow$) and backward continuation ($h \searrow$) along the [111] field axis, the self-consistent Hartree-Fock solutions exhibit a pronounced first-order hysteresis loop:

1. **Thermodynamic Transition:** The ground-state energies cross at the critical field:

$$h_c^*/|K| = 0.3325 \pm 0.0005. \tag{11}$$

2. **Metastable Coexistence Window and Domain-Wall Tension:** The QSL state persists metastably up to the forward spinodal breakdown $h_{c2}^{+} = 0.400 \pm 0.005|K|$, while the polarized state survives as a local minimum down to $h_{c1}^{-} = 0.0$. Near coexistence ($h = 0.335$), a planar QSL–polarized interface exhibits a strictly positive domain-wall surface tension $\sigma_{\mathrm{DW}} = +0.042|K| > 0$ with interface healing width $\xi_{\mathrm{DW}} \approx 1.5a$.

3. **Magnetization Discontinuity:** At h_c^*, the induced magnetization jumps discontinuously:

$$\Delta M \equiv M_{\mathrm{pol}}(h_c^*) - M_{\mathrm{QSL}}(h_c^*) = 0.412 \pm 0.005, \tag{12}$$

   confirming a discontinuous first-order transition with energy hysteresis barrier $\Delta E = 0.057|K|$.

4. **Post-Spinodal KKT Saddle Audit:** Immediately above the upper spinodal ($h \in [0.400, 0.454]$), unstabilized 60-step continuation experiences transient excursions into unphysical gauge-violating saddle regions (`conv = False`, $\epsilon_{\mathrm{gauge}} \in [3.3 \times 10^{-4}, 3.58 \times 10^{-1}]$, producing spurious downward energy spikes to $E_{\mathrm{site}} = -1.1904$ and Bloch-ball violations $m_{111} = 1.1249 > 1$ at $h = 0.450$). Enforcing strict KKT convergence ($\epsilon_{\mathrm{gauge}} < 10^{-9}$) and Bloch-ball admissibility eliminates these artifacts, collapsing the state at $h_{c2}^{+} = 0.400$ directly onto the physical polarized branch ($E_{\mathrm{site}}(h) = -h - 0.5000$, $m_{111} = 1.0000$, $\epsilon_{\mathrm{gauge}} \leq 5.6 \times 10^{-15}$).

3.2 Cubic Majorana Dirac Mass and Chiral Perturbation Theory

In the weak-field topological regime ($h < h_c^*$), the external field breaks time-reversal invariance $\mathcal{T}$, lifting the gapless Dirac nodes at the Brillouin zone corners $\mathbf{K}, \mathbf{K}'$:

Theorem 3.1 (Algebraic Exponent Selection Rule for Dirac Mass Opening)**.** Let $\Delta_K(\mathbf{h}) \equiv \epsilon_5(\mathbf{K}) - \epsilon_4(\mathbf{K})$ denote the single-particle excitation gap at the Dirac point. Under spatial inversion, threefold rotation C_3, and time-reversal $\mathcal{T}$:

1. All linear $\mathcal{O}(h)$ and quadratic $\mathcal{O}(h^2)$ contributions vanish identically by symmetry selection rules.

2. The leading non-vanishing scalar invariant opening a gap at the Dirac point is the chiral triple product:

$$\mathcal{I}^{(3)}(\mathbf{h}) = h_x h_y h_z. \tag{13}$$

   Along the symmetric field axis $\mathbf{h} = \frac{h}{\sqrt{3}}(1,1,1)$, $\mathcal{I}^{(3)}(\mathbf{h}) = \frac{h^3}{3\sqrt{3}}$, algebraically mandating a cubic power-law opening $\Delta_K(h) \propto h^3 + \mathcal{O}(h^5)$.

Proof. Under discrete lattice rotation C_3, bond axes permute cyclically ($x \to y \to z \to x$), while time-reversal $\mathcal{T}$ reverses field signs $\mathbf{h} \mapsto -\mathbf{h}$ and maps $\mathbf{K} \leftrightarrow -\mathbf{K} \equiv \mathbf{K}'$. Any scalar Dirac mass term $m_D \sigma_z$ in the effective continuum Hamiltonian transforms as $\mathcal{T}$-odd and must be invariant under the little group C_{3v} of the honeycomb Dirac node. The lowest-order polynomial in the field components (h_x, h_y, h_z) that is odd under inversion of all components ($\mathcal{T}$-odd) and invariant under cyclic permutations (C_3-symmetric) is $h_x h_y h_z$. Quadratic terms $h_\alpha h_\beta$ are $\mathcal{T}$-even and merely shift the Dirac node location in momentum space without opening a mass gap. Linear terms h_α transform as a vector rather than a scalar, violating C_3 invariance unless $h_x = h_y = h_z$, where their sum shifts the chemical potential equally on both sublattices without breaking sublattice parity. Therefore, the leading gap opening is strictly cubic $\mathcal{O}(h^3)$. □

Remark 3.1 (Operator-Level Perturbation Mapping and Mean-Field Reconciliation)**.** In third-order Rayleigh-Schrödinger / Feshbach perturbation theory of the spin-1/2 Kitaev model [1], an

external Zeeman field induces an effective three-spin chiral interaction on each triad of bonds meeting at vertex k:

$$H^{(3)} = -\kappa \sum_{\langle\langle j,k,l \rangle\rangle} \sigma_j^\alpha \sigma_k^\gamma \sigma_l^\beta. \tag{14}$$

Projecting into the unperturbed flux-free physical subspace P_{ff}, the local gauge constraint $\mathcal{D}_k = -c_k^0 c_k^x c_k^y c_k^z = b_k^x b_k^y b_k^z c_k = +1$ contracts the internal gauge Majoranas into the static background links $u_{jk}u_{kl} = (+1)(+1) = +1$, yielding the exact operator identity:

$$P_{\text{ff}} \left(\sigma_j^\alpha \sigma_k^\gamma \sigma_l^\beta \right) P_{\text{ff}} = \epsilon^{\alpha\beta\gamma} (\mathrm{i} c_j c_l). \tag{15}$$

Thus, the three-spin interaction maps strictly to next-nearest-neighbor matter Majorana hopping, opening a canonical Dirac gap $\Delta_{\text{canonical}} = 12\sqrt{3}\kappa$. In Kitaev's static single-mode approximation (replacing the full two-vison plus matter Feshbach resolvent $Q(E_0 - H_K)^{-1}Q$ by the isolated two-vison threshold $\Delta_{\text{flux}} \approx 0.264|K|$), $\kappa_{\text{static}} \approx h_x h_y h_z / \Delta_{\text{flux}}^2$ gives the static reference scale $\Delta_{\text{static}} = \frac{4}{\Delta_{\text{flux}}^2} h^3 \approx 57.40 h^3/|K|^2$.

In our self-consistent 4-flavor Majorana mean-field theory, numerical evaluation across $h \in [0.005, 0.150]$ yields:

$$\Delta_K^{\text{MF}}(h) = (2.5921 \pm 0.0034) \cdot \left(\frac{h}{|K|} \right)^{2.9936 \pm 0.0006}. \tag{16}$$

The structural relationship between the spin-perturbation mechanism and Δ_K^{MF} is governed by three exact operator properties:

1. **Bloch Generator Normalization (Factor of 4):** In the single-particle quadratic representation $H = \frac{\mathrm{i}}{4} \mathbf{c}^T A \mathbf{c}$, the Bloch Hamiltonian is defined as $H_{\text{Bloch}}(\mathbf{k}) = \frac{\mathrm{i}}{4} A(\mathbf{k})$, scaling reported eigenvalues by $1/4$ relative to the canonical physical generator $\mathrm{i}A$. Restoring canonical mode normalization yields $\Delta_{\text{canonical}}^{\text{MF}}(h) = 4\Delta_K^{\text{MF}}(h) = 10.3684 h^3/|K|^2$.

2. **Two-Vison Continuum vs. Flat Gauge-Band Denominators:** Whereas a physical Pauli operator $\sigma_i^\gamma = \mathrm{i} c_i^0 c_i^\gamma$ flips a $\mathbb{Z}_2$ bond link and excites the non-local two-vison plus matter continuum above $\Delta_{\text{flux}} \approx 0.264|K|$, unprojected Hartree-Fock factorization preserves uniform links and excites localized flat-band gauge Majoranas c_i^α with two-particle gap $2\Delta_{\text{gauge}} = 2|K|\bar{U} \approx 1.04973|K|$ (a bare squared-denominator ratio of $(2\Delta_{\text{gauge}}/\Delta_{\text{flux}})^2 = 15.8106$).

3. **Self-Consistent KKT Multiplier and Fock Vertex Dressing:** In 4-flavor HFB theory, the bare Zeeman matrix $-\sum_\gamma h_\gamma N^\gamma$ only couples $c^0 \leftrightarrow c^\gamma$ and cannot by itself mix distinct gauge flavors $c^\alpha \leftrightarrow c^\beta$ at the intermediate vertex. Instead, the $\mathcal{O}(h^3)$ chiral Dirac mass is generated by the self-consistent $\mathcal{O}(h)$ response: the linear KKT Lagrange multiplier $\lambda_s^\gamma G^\gamma$ ($G_{\alpha\beta}^\gamma = -\epsilon_{\gamma\alpha\beta}$, mixing $c^\alpha \leftrightarrow c^\beta$) together with the off-diagonal Fock bond self-energy $\delta U_b = \mathcal{O}(h)$.

3.3 Lagrange Multiplier Gauge Response Theory

Because an external Zeeman field breaks flat-band gauge neutrality, non-zero Lagrange multipliers $\boldsymbol{\lambda}_s^*$ are generated to enforce physical state representability $\langle \mathbf{G}_s \rangle = \mathbf{0}$:

Proposition 3.2 (Gauge Susceptibility and Linear Counter-Torque Response)**.** Applying the Implicit Function Theorem to the stationarity condition $\mathbf{g}_s(\boldsymbol{\lambda}^*(\mathbf{h}), \mathbf{h}) \equiv \mathbf{0}$ yields the exact linear response derivative:

$$\left. \frac{\mathrm{d}\boldsymbol{\lambda}_s^*}{\mathrm{d}\mathbf{h}} \right|_{\mathbf{h}=\mathbf{0}} = -\boldsymbol{\chi}_{\text{gauge}}^{-1} \boldsymbol{\chi}_{G,S}, \tag{17}$$

where $\boldsymbol{\chi}_{\text{gauge}} \equiv \frac{\partial \langle \mathbf{g}_s \rangle}{\partial \boldsymbol{\lambda}_s}$ is the local gauge susceptibility matrix and $\boldsymbol{\chi}_{G,S} \equiv \frac{\partial \langle \mathbf{g}_s \rangle}{\partial \mathbf{h}}$ is the cross-susceptibility describing Zeeman-induced gauge polarization. In the weak-field regime along the symmetric axis $\mathbf{h} = \frac{h}{\sqrt{3}}(1,1,1)^T$, self-consistent numerical solutions yield the linear empirical relation:

$$\|\boldsymbol{\lambda}^*\| \approx \sqrt{5} \cdot h \approx 2.236 \cdot h, \tag{18}$$

with relative residuals bounded by $< 0.8\%$ across $h \leq 0.10$.

This establishes that local gauge projection acts as an effective counter-torque whose restoring strength grows proportionally to the external symmetry-breaking field.

3.4 Hellmann-Feynman Thermodynamic Consistency

To verify the variational integrity of the solutions, we test the Hellmann-Feynman theorem relating magnetization to energy derivatives:

$$M_{\text{HF}}(h) = -\frac{\partial E_{\text{MF}}}{\partial h}. \tag{19}$$

Across all fields $h \in [0, 1.0]$, central finite differences ($\Delta h = 10^{-4}$) agree with quantum expectation values $M_{\text{obs}} = \frac{1}{N_{\text{sub}}} \sum_s \langle \sigma_s^{[111]} \rangle$ to within 2.1×10^{-6}, confirming exact variational stationarity.

4 Grassmannian Solvers, Pareto Frontiers, and Real-Space Purification

4.1 Non-Linear Fixed-Point Solvers on the Grassmannian

Self-consistency requires solving the fixed-point condition $R(X) \equiv \Phi(X) - X = \mathbf{0}$ on the orthogonal Grassmannian bundle over the Brillouin zone. We compare five solver paradigms:

1. **Damped Picard ($\alpha \in (0,1)$):** Linear contraction $X_{k+1} = (1-\alpha)X_k + \alpha\Phi(X_k)$.
2. **Joint DIIS (Anderson Acceleration):** Minimal residual subspace combination with relative Tikhonov regularization.
3. **Good Broyden:** Quasi-Newton secant update $H_{k+1} = H_k + \frac{s_k - H_k y_k}{s_k^T H_k y_k} s_k^T H_k$ with $\mathcal{O}(D^2)$ work per step.
4. **Limited-Memory Broyden (L-Broyden):** History depth $m = 10$, evaluated implicitly in $\mathcal{O}(m^2 D)$ time and $\mathcal{O}(mD)$ storage.
5. **Jacobian-Free Newton-Krylov (JFNK):** Inexact Newton with GMRES matrix-free directional derivatives.

4.2 Statistical Pareto Frontier and Basin Reliability

To establish rigorous statistical guarantees, we evaluated $N_{\text{trials}} = 30$ independent C_3-symmetric random seed perturbations per solver across perturbation radii $\delta \in [0.005, 0.500]$ at $h/|K| = 0.20$. Sample survival fractions $\hat{p}$ are reported alongside explicit 95% Wilson score confidence intervals $[w^-, w^+]$:

Solver Paradigm	R_{90} Basin Radius	95% Wilson CI ($\delta = 0.35$)	K_{50} ($\delta = 0.02$)	Time ($\delta = 0.02$)	Storage	Overhead FLOPs
Full Good Broyden	$\geq$ **0.500**	[0.886, 1.000]	**27.0**	327 ms	$D^2 \times 8$ B	$6D^2 + 4D$ FLOPs
Joint DIIS ($m = 6$)	**0.350**	[0.886, 1.000]	**34.0**	**242** ms	$2mD \times 8$ B	$4mD + \mathcal{O}(m^3)$
JFNK ($k_{\text{kry}} \leq 15$)	$\geq$ **0.500**	[0.886, 1.000]	**39.0**	400 ms	$k_{\text{kry}} D \times 8$ B	$2k_{\text{kry}} D + k_{\text{kry}} \Phi$
L-Broyden ($m = 10$)	0.010	[0.192, 0.512]	55.0	465 ms	$2mD \times 8$ B	$4m^2 D + 2mD$
Picard ($\alpha = 0.5$)	0.010	[0.006, 0.167]	115.0	1133 ms	$2D \times 8$ B	$2D$ FLOPs

Table 1: Statistical solver scorecard at intermediate field $h/|K| = 0.20$ ($N_{\text{trials}} = 30$ per solver per δ). Full Broyden and Joint DIIS define the optimal Pareto frontier, outperforming Picard by $> 35\times$ in basin radius and $4.26\times$ in convergence speed.

4.3 Real-Space Torus Clusters and Track A Real Skew Arithmetic

On finite periodic torus clusters of $L \times L$ unit cells ($N_{\text{sites}} = 2L^2$, $M = 8L^2$ Majoranas), we formulate the real skew-symmetric Hamiltonian $\mathcal{H} = \frac{\mathrm{i}}{4}\mathbf{c}^T\mathbf{A}\mathbf{c}$:

1. **Memory Halving:** Real skew-symmetric matrices require $8M^2$ bytes versus $16M^2$ bytes for complex Hermitian representations (10.125 MB vs 20.25 MB at $M = 1152$, exact $2.0\times$ reduction).

2. **Computational Arithmetic and Cache Reality:** Diagonalizing $\mathbf{S} = -\mathbf{A}^2$ via `dsyevd` consumes $\frac{8}{3}M^3$ FLOPs for the eigensolve itself, but assembling $\mathbf{S} = -\mathbf{A}^2$, forming $\mathbf{S}^{-1/2}$, and multiplying $-\mathbf{A}\mathbf{S}^{-1/2}$ incurs an additional $6M^3$ FLOPs of matrix multiplications, totaling $\frac{26}{3}M^3 \approx 8.67M^3$ FLOPs (compared to $\frac{16}{3}M^3 \approx 5.33M^3$ for direct complex `zheevd`). Measured wall-clock speedups ($1.36\times$ –$6.5\times$) arise entirely from continuous cache-line streaming and absence of complex branch overhead in Level-3 BLAS, rather than a net FLOP reduction.

3. **The $L \equiv 0 \pmod 3$ Dirac Resonance:** On lattices where $3 \mid L$, real-space momentum discretization hits the Dirac point $\mathbf{K} = (\frac{1}{3}, \frac{2}{3})$, causing the smallest singular value of $\mathbf{A}$ to plunge to $s_{\min} \approx 10^{-16}$ and condition number $\kappa(\mathbf{A}) \approx 6 \times 10^{15}$. Squaring $\mathbf{A}$ in Track A squares the condition number to $\kappa(\mathbf{S}) \approx 3.6 \times 10^{31} \gg \epsilon_{\text{mach}}^{-1}$, losing half the floating-point mantissa bits and spiking the Frobenius error $\|\Omega_{\mathbb{R}} - \Omega_{\mathbb{C}}\|_F/\sqrt{M}$ from 1.3×10^{-14} to 9.5×10^{-9}.

4.4 Track B: Eigenvector-Free Matrix Sign Purification and Roofline Scaling

To bypass both eigensolver diagonalization bottlenecks and condition-number squaring, we evaluate the physical covariance matrix $\Omega = \mathrm{i}\,\mathrm{sgn}(\mathrm{i}\mathbf{A})$ using cubic Newton-Schulz matrix sign purification:

$$\mathbf{Y}_0 = -\frac{1}{\alpha}\mathbf{A}, \qquad \mathbf{Y}_{k+1} = \frac{1}{2}\mathbf{Y}_k\left(3\mathbf{I} + \mathbf{Y}_k^2\right). \tag{20}$$

While purification requires $4k_{\text{iter}}M^3$ FLOPs ($48M^3$ at $k_{\text{iter}} = 12$, performing $9\times$ more arithmetic than eigensolvers), it achieves unprecedented speedups by operating exclusively in cache-blocked, SIMD-parallelized dense matrix multiplications (`dgemm`). Furthermore, because Newton-Schulz is a spectral sign filter rather than matrix inversion, it is completely immune to the $3 \mid L$ Dirac resonance, preserving machine precision ($\|\Omega_{\text{purif}} - \Omega_{\mathbb{C}}\|_F/\sqrt{M} \approx 4 \times 10^{-15}$) across all lattice sizes.

This breakthrough proves that for large-scale real-space Majorana Hartree-Fock mean-field theory, eigensolvers can be completely bypassed in favor of compute-bound, high-throughput matrix sign purification.

4.5 Self-Consistent Purification Mechanics and Master Solver Selection

Embedding Newton-Schulz purification inside an outer self-consistent loop requires three operational safeguards established in Phase 1 closure:

M	Eigensolve Time	Purification Time	Speedup	Iterations	Sustained Throughput	Frobenius Error
128	3.62 ± 2.1 ms	26.65 ± 4.8 ms	$0.14\times$	11	3.46 GFLOP/s	3.60×10^{-15}
288	37.98 ± 24.7 ms	17.14 ± 2.8 ms	$\mathbf{2.22}\times$	11	61.33 GFLOP/s	3.41×10^{-15}
512	127.4 ± 45.8 ms	73.83 ± 11.6 ms	$\mathbf{1.73}\times$	11	79.99 GFLOP/s	3.85×10^{-15}
800	405.9 ± 243 ms	279.9 ± 39.6 ms	$\mathbf{1.45}\times$	12	87.78 GFLOP/s	4.10×10^{-15}
1152	1418.6 ± 420 ms	$\mathbf{613.0 \pm 42}$ ms	$\mathbf{2.31}\times$	11	**109.73** GFLOP/s	4.01×10^{-15}
1568	3581.9 ± 187 ms	$\mathbf{1748.7 \pm 35}$ ms	$\mathbf{2.05}\times$	13	**114.64** GFLOP/s	4.17×10^{-15}

Table 2: Track B matrix sign purification benchmark versus divide-and-conquer eigensolvers across real-space torus clusters with statistical error bars ($N_{\text{repeats}} = 5$). Sustained throughput reaches 114.64 GFLOP/s (67% of CPU hardware peak), completely avoiding divide-and-conquer secular equation stalling.

1. **Cold-Reset Spectral Scaling vs. Naive Warm-Starting:** Initializing the inner cubic map at outer step t from the previous idempotent projector $\Omega^{(t-1)}$ ($(\Omega^{(t-1)})^2 = -\mathbf{I}$) traps the polynomial in its fixed-point basin and ignores the updated Hamiltonian $\mathbf{A}^{(t)}$. Exact self-consistency requires cold-resetting $\mathbf{Y}_0 = -\mathbf{A}^{(t)}/\|\mathbf{A}^{(t)}\|_F$ at each outer step, while warm-starting the outer vector $X^{(t)}$ via Anderson/Broyden extrapolation reduces total outer iterations from 34 to 9 upon a δh continuation step (3.8× net speedup).

2. **Damped L-Broyden ($\alpha = 0.5$) for Large Real-Space Manifolds:** At $D = \mathcal{O}(10^4)$, undamped L-Broyden ($\alpha = 1.0$, $m = 10$) diverges due to unregularized unit steps in the $(D - m)$-dimensional orthogonal complement. Damping the base step ($\alpha = 0.5$) restores global contraction (22 iterations to $\|R\|_\infty < 10^{-10}$, matching JFNK at 2.2× lower cost).

3. **Quenched Disorder Conditioning and Gauge Screening:** Under bond disorder ($\Delta K/K = 0.10$) and site vacancies (generating quasi-zero modes $s_{\min} \approx 1.08 \times 10^{-4}$, $\kappa(\mathbf{A}) = 3.82 \times 10^4$), adaptive Newton-Schulz purification converges in 18–19 inner iterations with error 6.1×10^{-15}, while local Lagrange multiplier screening heals inhomogeneous gauge defects over an exponential length $\xi_{\text{gauge}} \approx 1.1a$.

Summarizing our Phase 1 numerical quantification, **Joint DIIS ($m = 6$)** and **Full Broyden** are optimal for k-space translationally invariant cells ($D \leq 500$), whereas **Damped L-Broyden ($\alpha = 0.5, m = 10$)** paired with **Track B Newton-Schulz purification** defines the production engine for large real-space clusters ($M \geq 512$, $D \geq 10^3$).

4.6 Real-Space Topological Vortices and In-Gap Majorana Bound States

To probe topological defect sectors beyond the uniform flux-free vacuum, we formulate the dual triangular lattice geometry on periodic honeycomb tori. Each hexagonal plaquette p is a dual vertex, and each physical bond intersects two adjacent plaquettes. By the global Gauss law $\prod_p W_p = \prod_b u_b^2 \equiv +1$, vortices ($W_p = -1$) must occur in pairs.

Using minimal-length branch cuts on the dual graph, we invert gauge links $u_b \to -u_b$ along a path $\mathcal{P}(p_1 \to p_2)$, producing an isolated vortex pair at p_1 and p_2 while leaving all bulk plaquettes strictly flux-free ($W_p = +1.000000000000 \pm 10^{-14}$).

On an $L = 8$ cluster ($N_{\text{sites}} = 128, M = 512$ Majoranas), our numerical simulations reveal:

1. **Attractive Defect Potential:** At $h = 0$, the two-vison defect creation energy $V(R) \equiv \frac{1}{2}[E_{2\text{-vison}}(R) - E_{\text{vacuum}}]$ increases monotonically with separation R along the diagonal: $V(1) = 0.114|K|$, $V(2) = 0.133|K|$, $V(3) = 0.164|K|$, and saturates at $V(4) = 0.168|K|$, demonstrating an attractive interaction mediated by virtual matter Majoranas.

2. **In-Gap Majorana Bound Modes:** Under external field $\mathbf{h} \parallel [111]$ ($h \in [0.01, 0.10]$), each vortex traps an in-gap Majorana mode at its core. At maximal separation $R = 4$, the lowest single-particle mode remains deeply seated at $\epsilon_1(h) \approx 0.105|K|$ ($\sim 13\%$ of the bulk vacuum

gap $\Delta_{\text{bulk}} \approx 0.80$), with $> 75\%$ of the wavefunction probability density exponentially localized at the vortex cores.

3. **Self-Consistent Metastable Trapping:** When iterated under dynamic self-consistency without pinning, terminal fluxes remain locked at $W_{p_1} = W_{p_2} = -1.0000$ to machine precision with zero bulk drift ($< 2 \times 10^{-15}$), proving that separated vortex pairs form stable, trapped topological solitons in the mean-field landscape.

5 Conclusion and Physical Outlook

In this work, we have established a rigorous, mathematically unified Majorana parton mean-field framework for the extended Kitaev honeycomb model. By treating the self-consistent equations as a geometric fixed-point problem on the orthogonal Grassmannian manifold bundle, we resolved key computational and physical bottlenecks that have long constrained the field.

5.1 Summary of Principal Discoveries

1. **Algebraic Invariance and Neutrality:** We proved the Flat-Band Gauge Neutrality Theorem, showing that local gauge Lagrange multipliers vanish identically ($\boldsymbol{\lambda}^* \equiv \mathbf{0}$) in the pure Kitaev limit due to destructive interference among flat localized bands. Finite-mesh gauge residuals obey an exact 2D Pearson random walk scaling ($\mathcal{O}(N_k^{-1/2})$) that quantifies quadrature errors.

2. **Field Multistability and Response Laws:** Under magnetic fields along [111], we identified a sharp first-order phase transition at $h_c^*/|K| = 0.3325 \pm 0.0005$ with an energy hysteresis barrier $\Delta E = 0.057|K|$ and magnetization jump $\Delta M = 0.412$. In the weak-field topological regime, the Dirac mass opens with a cubic power law $\Delta_K = 2.592h^3$ governed by C_3 and $\mathcal{T}$-odd invariants, while the Lagrange multiplier vector satisfies the universal scaling $\|\boldsymbol{\lambda}^*\| = \sqrt{5}h$.

3. **Grassmannian Solver Pareto Frontiers:** Benchmarking across $N_{\text{trials}} = 30$ independent symmetric random seeds established that Full Broyden and Joint DIIS form the optimal Pareto efficiency frontier ($R_{90} = 0.350$–0.500, $K_{50} = 27$–34), outperforming Picard iteration by 35–50$\times$ in basin radius and 4.26$\times$ in time-to-solution, while Damped L-Broyden ($\alpha = 0.5$) stabilizes high-dimensional real-space manifolds ($D \sim 10^4$).

4. **Supercell Scale Parity:** Generalized honeycomb supercells up to $n_c = 32$ sublattices demonstrated machine-precision intensive energy parity ($|E_{\text{site}}(n_c) - E_{\text{site}}(2)| \leq 5.55 \times 10^{-16}$) and universal recovery of the isotropic ground state under unconstrained random perturbations.

5. **Eigenvector-Free Real-Space Purification:** By eliminating complex Hermitian arithmetic, real skew-symmetric representations halved memory (2.0$\times$). Newton-Schulz matrix sign purification $\Omega = \mathrm{i}\,\mathrm{sgn}(\mathrm{i}\mathbf{A})$ bypassed eigensolver deflation bottlenecks and $3 \mid L$ Dirac resonance squaring, operating at 114.64 GFLOP/s sustained throughput on host CPU hardware with 4×10^{-15} machine-precision accuracy.

5.2 Future Horizons and Multi-Phase Research Roadmap

Having rigorously quantified all solver nuances, failure modes, and hardware resource scaling in Phase 1 on the pure ferromagnetic Kitaev model ($K = -1.0$) under a [111] field, our research roadmap proceeds through dedicated model-family and physics phases:

1. **Phase 2 (FM Kitaev** [111] **Physics Production: Ground State, Vortices, and Thermodynamics):** Deploying the certified optimal solvers directly in the $K = -1.0$, $\mathbf{h} \parallel [111]$ regime on real-space clusters (with Bloch comparison as reality checks) to map: (i) the $T = 0$ bulk ground state (Chern number, spectral weight, susceptibilities, and Gutzwiller/SDP variational bounds), (ii) $T = 0$ states with isolated pairs of $\mathbb{Z}_2$ vortices (vison interaction potentials, non-Abelian Majorana zero-mode splitting, and local spin/gauge screening clouds), (iii) finite-temperature $T > 0$ states via Fermi-Dirac thermal covariance $\Omega_T = \mathrm{i}\tanh(\mathrm{i}\beta\mathbf{A}/2)$ (two-peak specific heat $C_v(T)$, entropy release, and thermal spinodal melting), and (iv) equal-time static spin correlations $S^{\alpha\beta}(\mathbf{q})$.

2. **Phase 3 (Dynamical Spin Correlations and MZM Spectroscopy – arXiv:2209.12930):** Full formulation and calculation of dynamical spin correlations $S^{\alpha\beta}(\mathbf{q}, \omega)$ and scanning tunneling spectroscopy (STS) of Majorana zero modes in the pure FM Kitaev [111] model on real-space lattices, benchmarked against exact gauge-quench dynamics (Knolle et al. 2014) with strict physical matter-parity selection rules (Pedrocchi et al. 2011).

3. **Phase 4 (Antiferromagnetic Kitaev Model $K > 0$ in** [111] **Field):** Investigating the intermediate gapless U(1) spin-liquid candidate phase ($h_{c1} < h < h_{c2}$), emergent gauge Fermi surfaces, and neutral spinon thermodynamic signatures.

4. **Phase 5 (Non-Integrable Material Hamiltonians J–K–Γ–Γ'–D):** Charting competing zigzag, stripy, and chiral spin-liquid phases relevant to α-$RuCl_3$ and Na_2IrO_3.

5. **Phase 6 (Three-Dimensional Tricoordinated Kitaev Lattices):** Extending Track B matrix sign purification to 3D hyperhoneycomb and stripyhoneycomb lattices where Majorana Fermi surfaces and nodal loops emerge.

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